Statistical Mechanics rediscovered under the lense of entropy maximization principles

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Contents

I Statistical Derivation of Ensembles **4**

1 Entropy Maximization Principle **5**

1.1 Gibbs-Shannon Entropy 5

1.2 Entropy Maximization 6

2 The Microcanonical Ensemble **7**

3 The Canonical Ensemble **8**

3.1 First and Second Law of Thermodynamics - Making Sense of Multipliers 10

3.2 Extensive and Intensive Variables 11

3.3 Energy Fluctuation, Thermodynamic Limit, and Equivalence of Ensembles 12

3.4 Zeroth Law of Thermodynamics 13

4 Beyond Canonical Ensemble **15**

4.1 Internal Energy Minimization 15

4.2 Thermodynamic Potential 16

4.3 The Legendre Transformation . . . 16

4.4 Multivariable Legendre Transformation . . . 17

4.5 NVT Ensemble - Helmholtz Free Energy . . . 18

4.6 NPS Ensemble - Enthalpy . . . 19

5 The Grand Canonical Ensemble **21**

5.1 Particle Number Fluctuation, Thermodynamic Limit, and Equivalence of Ensembles (again) . . . 23

5.2 μVT Ensemble - Grand Potential . . . 23

5.3 NPT Ensemble - Gibbs Free Energy . . . 24

5.4 Euler's Homogeneous Function Theorem . . . 25

5.5 Multi-Component System . . . 26

5.6 Mixing Entropy and the Gibbs Paradox . . . 28

5.7 Gibbs-Duhem Relation . . . 29

II Phase Transtion **31**

6 First-Order Phase Transitions and Phase Diagram **32**

6.1 Gibbs Phase Rule . . . 32

6.2 Clapeyron Equation and Fixed-Composition (P–T) Phase Diagram . . . 33

6.3 Unary System . . . 34

7 Binary System **35**

7.1 The X–P–T Diagram . . . 35

III Classical thermodnynamic **36**

IV Physical Chemistry **36**

V Quantum Statistical Mechanics **36**

Part I

Statistical Derivation of Ensembles

1 Entropy Maximization Principle

The conventional approach and its limitation. The conventional introduction to statistical mechanics rests on four notions:

- A microstate is a complete description of every microscopic parameter of the system.
- In an isolated system, all microstates are equally likely.
- A macrostate is a group of microstates that share the same macroscopic observables.
- Macrostates that contain more microstates are observed more often, so the observed macrostate is the most probable one.

Boltzmann, Gibbs, and other founders of thermodynamics developed this approach. It is physically intuitive, but explaining why a system settles into the most probable macrostate requires ergodicity and dynamical arguments, which are difficult to make rigorous.

The MaxEnt alternative. In this document, we follow Jaynes and treat equilibrium statistical mechanics as the solution of a constrained optimization problem: maximize the entropy subject to what is macroscopically known about the system. We call this the Entropy Maximization Principle (MaxEnt).

1.1 Gibbs-Shannon Entropy

From "disorder" to ignorance. The meaning of entropy has long been debated. The popular description of entropy as "disorder" is not a definition, because it gives no quantitative way to measure disorder. Clausius, Boltzmann, and other founders regarded entropy as a driving force of thermodynamic processes, which left open the question of which physical quantity entropy encodes. Jaynes proposed instead that entropy is not a property of the body itself, but a measure of our ignorance about it. The Gibbs-Shannon entropy formalizes this idea for Ω events that occur with probabilities $p(i)$:

$$\boxed{S(p) = -\sum_{i=1}^{\Omega} p(i) \ln p(i).} \tag{1.1}$$

Why the Gibbs-Shannon entropy measures ignorance. The term $-\ln p(i)$ measures our ignorance about event i, also called its surprisal:

- A certain event has $p(i) = 1$, so its ignorance is $-\ln 1 = 0$.
- A rare event has small $p(i)$, so its ignorance is large.

Summing $-\ln p(i)$ directly would fail, because knowing that an event will occur and knowing that it will not occur both mean low ignorance. We therefore weight each term by $p(i)$. Rare events seldom occur, and near-certain events are already expected, so both contribute little to the total. Equivalently, S is the expected value of $-\ln p$.

1.2 Entropy Maximization

The principle of maximum ignorance. Jaynes' second idea is that, without knowledge of the underlying mechanism, we should choose the distribution that maximizes our ignorance, that is, the Gibbs-Shannon entropy $S(p)$, subject to the known constraints. A lower-entropy distribution would introduce an unexplained bias toward some events. For example, a coin flip should assign equal probabilities to both outcomes unless a constraint, such as a loaded coin, states otherwise.

Formulating the constrained optimization. For simplicity, we write p_i instead of $p(i)$ from now on, as there is no need to distinguish between different probability distributions. We write each known constraint as $g_a(p) = 0$. For the expectation value of an observable with value $g_a(i)$ in event i,

$$g_a(p) = \sum_i p_i \, g_a(i) - \langle g_a \rangle.$$

Lagrange multipliers remove the constraints. We define the Lagrangian

$$\boxed{\mathcal{L}(p) = -\sum_i p_i \ln p_i - \alpha \left(\sum_i p_i - 1 \right) - \sum_a \lambda_a \, g_a(p),} \tag{1.2}$$

where α enforces normalization and each λ_a enforces one constraint. The maximizing distribution satisfies

$$\boxed{\frac{\partial \mathcal{L}}{\partial p_i} = 0, \qquad \frac{\partial \mathcal{L}}{\partial \alpha} = 0, \qquad \frac{\partial \mathcal{L}}{\partial \lambda_a} = 0.} \tag{1.3}$$

The last two conditions reproduce normalization and the constraints. The first condition determines p_i.

The general MaxEnt distribution. For expectation-value constraints, $\partial g_a / \partial p_i = g_a(i)$. The first condition in Eq. (1.3) then gives

$$-\ln p_i - 1 - \alpha - \sum_a \lambda_a \, g_a(i) = 0,$$

$$p_i = e^{-1-\alpha} \exp\Big(-\sum_a \lambda_a \, g_a(i) \Big).$$

Normalization fixes α and yields

$$\boxed{p_i = \frac{1}{Z} \exp\Big(-\sum_a \lambda_a \, g_a(i) \Big), \qquad Z = \sum_i \exp\Big(-\sum_a \lambda_a \, g_a(i) \Big).} \tag{1.4}$$

Since S is strictly concave in p and the constraints are linear, this stationary point is the unique maximum.

2 The Microcanonical Ensemble

The Boltzmann constant in the entropy. In thermodynamics, the Gibbs-Shannon entropy carries the Boltzmann constant k_B, which sets the units of entropy:

$$\boxed{S(p) = -k_B \sum_{i=1}^{\Omega} p_i \ln p_i.} \tag{2.1}$$

For simplicity, we set $k_B = 1$ during derivations and restore it when stating final results.

From normalization alone to Boltzmann's entropy. We first consider the simplest case, in which the only constraint is normalization, $\sum_i p_i = 1$. This case describes the microcanonical ensemble: an isolated system with fixed energy and other fixed macroscopic constraints. The energy is not imposed through a Lagrange multiplier. Instead, it restricts the sample space to the microstates with energy U, in practice a thin shell $[U, U + \delta U]$. Setting the derivatives of the Lagrangian in Eq. (1.3) to zero gives

$$\frac{\partial \mathcal{L}}{\partial p_i} = -\ln p_i - 1 - \alpha = 0, \qquad \frac{\partial \mathcal{L}}{\partial \alpha} = -\Big(\sum_i p_i - 1\Big) = 0.$$

The first condition gives $p_i = e^{-1-\alpha}$, which does not depend on i. Normalization then requires $\Omega\, e^{-1-\alpha} = 1$, so

$$\boxed{p_i = \frac{1}{\Omega}.} \tag{2.2}$$

Maximizing the entropy under normalization alone therefore yields a uniform distribution: all accessible microstates are equally probable. The conventional approach postulates this property. Here, MaxEnt derives it, and it recovers Boltzmann's premise for the microcanonical ensemble. Substituting $p_i = 1/\Omega$ into the entropy gives

$$S = -\sum_{i=1}^{\Omega} \frac{1}{\Omega} \ln \frac{1}{\Omega} = \ln \Omega,$$

and restoring k_B yields Boltzmann's microcanonical entropy:

$$\boxed{S = k_B \ln \Omega.} \tag{2.3}$$

Dependence of the entropy on the macroscopic constraints. The number of accessible microstates, Ω, is also called the microcanonical partition function. It depends on the system's Hamiltonian $\mathcal{H}$ and on the macroscopic constraints. In particular, the accessible microstates are set by the volume V, the particle number N, and the internal energy U, so the entropy is a function of these three variables:

$$\boxed{S = S(N, V, U).} \tag{2.4}$$

Specifying N, V, and U therefore determines the equilibrium entropy uniquely.

3 The Canonical Ensemble

Constraining the average of an observable. The microcanonical ensemble does not distinguish between accessible microstates: only their number matters. Realistic systems, however, obey constraints imposed by measurement. Suppose microstate i has a definite value a_i of an observable A, and the measured average is $\bar{A}$. The expectation value is

$$\boxed{\langle A\rangle \equiv \sum_i p_i a_i,} \tag{3.1}$$

and the constraint reads $\langle A\rangle = \bar{A}$. Here, $\langle A\rangle$ denotes the operation that extracts the mean of the observable A, and $\bar{A}$ denotes the numerical value assigned to that mean. Since only one constraint is present, we also rename the multiplier λ_a as β. The Lagrangian now contains the multiplier α for normalization and the multiplier β for the constraint on $\langle A\rangle$. Its stationarity conditions are

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial p_i} &= -\ln p_i - 1 - \alpha - \beta a_i = 0, \\
\frac{\partial \mathcal{L}}{\partial \alpha} &= -\Big(\sum_i p_i - 1\Big) = 0, \\
\frac{\partial \mathcal{L}}{\partial \beta} &= -\Big(\sum_i a_i p_i - \bar{A}\Big) = 0.
\end{aligned}$$

The Boltzmann distribution and the partition function. The first condition gives $p_i = e^{-1-\alpha-\beta a_i}$. Substituting this into normalization and into the constraint yields

$$e^{-1-\alpha}\sum_i e^{-\beta a_i} = 1, \qquad e^{-1-\alpha}\sum_i a_i e^{-\beta a_i} = \bar{A}.$$

The first equation determines α in terms of β and leaves a single free parameter. Defining the canonical partition function

$$\boxed{Z(\beta) \equiv \sum_i e^{-\beta a_i},} \tag{3.2}$$

the probability distribution becomes

$$\boxed{p_i = \frac{e^{-\beta a_i}}{Z(\beta)}.} \tag{3.3}$$

This is the Boltzmann distribution. For $\beta > 0$, states with larger a_i are less probable. The remaining constraint, $\langle A\rangle(\beta) = \bar{A}$, determines β.

Uniqueness of the multiplier. To show that β is unique, we first express the mean through Z:

$$\langle A\rangle = \frac{1}{Z}\sum_i a_i e^{-\beta a_i} = -\frac{\partial \ln Z}{\partial \beta}.$$

Hence

$$\boxed{\langle A\rangle = -\frac{\partial \ln Z}{\partial \beta}.} \tag{3.4}$$

Differentiating once more, with $\mathrm{Var}(A) \equiv \langle A^2\rangle - \langle A\rangle^2$,

$$\begin{aligned}\frac{\partial\langle A\rangle}{\partial\beta} &= \frac{\partial}{\partial\beta}\left(\frac{\sum_i a_i e^{-\beta a_i}}{Z}\right)\\ &= -\frac{\sum_i a_i^2 e^{-\beta a_i}}{Z} + \frac{1}{Z^2}\left(\sum_i a_i e^{-\beta a_i}\right)^2\\ &= -\langle A^2\rangle + \langle A\rangle^2 = -\mathrm{Var}(A),\end{aligned}$$

so

$$\boxed{\frac{\partial\langle A\rangle}{\partial\beta} = -\mathrm{Var}(A).} \tag{3.5}$$

Since $\mathrm{Var}(A) \geq 0$, $\langle A\rangle(\beta)$ is monotonically non-increasing. It is strictly decreasing whenever A is not identical across all accessible microstates. Thus, for target values in the interior of the achievable range, $\langle A\rangle$ determines β uniquely.

Entropy of the canonical distribution and the meaning of β. We now compute the entropy of the resulting distribution, with k_B omitted during the derivation:

$$\begin{aligned}S &= -\sum_i p_i \ln p_i = -\sum_i p_i\left(-\beta a_i - \ln Z\right)\\ &= \beta\sum_i a_i p_i + \ln Z \sum_i p_i = \beta\langle A\rangle + \ln Z.\end{aligned}$$

Restoring k_B,

$$\boxed{S = k_B\left(\beta\langle A\rangle + \ln Z\right).} \tag{3.6}$$

Although S depends on $\bar{A}$ both explicitly and implicitly through $\beta(\bar{A})$, differentiating gives

$$\begin{aligned}\frac{\partial S}{\partial\langle A\rangle} &= \beta + \langle A\rangle\frac{d\beta}{d\langle A\rangle} + \frac{\partial \ln Z}{\partial\beta}\frac{d\beta}{d\langle A\rangle}\\ &= \beta + \langle A\rangle\frac{d\beta}{d\langle A\rangle} - \langle A\rangle\frac{d\beta}{d\langle A\rangle} = \beta.\end{aligned}$$

Restoring k_B yields the meaning of the multiplier:

$$\boxed{\beta = \frac{1}{k_B}\frac{\partial S}{\partial\langle A\rangle}.} \tag{3.7}$$

Concavity of the entropy. A function f is concave on an interval I if, for all $x, y \in I$ and all $t \in [0,1]$,

$$\boxed{f\big(tx + (1-t)y\big) \;\geq\; t\,f(x) + (1-t)\,f(y).} \tag{3.8}$$

Geometrically, every chord lies on or below the graph. The function is strictly concave if the inequality is strict for $x \neq y$ and $t \in (0,1)$. For a twice-differentiable f, concavity is equivalent to $f'' \leq 0$, and $f'' < 0$ guarantees strict concavity.

We apply this criterion to S as a function of $\langle A \rangle$. Using the inverse-function rule and $\partial\langle A\rangle/\partial\beta = -\mathrm{Var}(A)$,

$$\frac{\partial^2 S}{\partial \langle A\rangle^2} = \frac{d\beta}{d\langle A\rangle} = \left(\frac{d\langle A\rangle}{d\beta}\right)^{-1} = -\frac{1}{\mathrm{Var}(A)}.$$

Restoring k_B,

$$\boxed{\frac{\partial^2 S}{\partial \langle A\rangle^2} = -\frac{k_B}{\mathrm{Var}(A)} < 0.} \tag{3.9}$$

The entropy is therefore strictly concave in $\langle A \rangle$ whenever A is not identical across all accessible microstates. This is the monotonic decrease of $\langle A\rangle(\beta)$ read from the side of S: each value of $\langle A\rangle$ corresponds to exactly one β.

Specialization to the internal energy. We have not specified the observable A, so the derivation applies to any observable whose expectation value is constrained. The canonical ensemble conventionally fixes the volume V and particle number N and constrains the internal energy. From here on, we take $A = U$ and write U_i for a_i. The results become

$$\boxed{\begin{aligned} Z(\beta) &= \sum_i e^{-\beta U_i}, \quad p_i = \frac{e^{-\beta U_i}}{Z(\beta)}, \\ \langle U\rangle &= -\frac{\partial \ln Z}{\partial \beta}, \qquad S = k_B\left(\beta\langle U\rangle + \ln Z\right). \end{aligned}} \tag{3.10}$$

We identify β with the temperature in Section 3.1. The reason for not constraining only the particle number or the volume will be discussed shortly.

Recovering the microcanonical ensemble. If every accessible microstate has the same energy U, the canonical ensemble reduces to the microcanonical one. All weights $e^{-\beta U}$ are identical, so the distribution is uniform and the mean energy is the exact value U. Then

$$\begin{aligned} Z &= \sum_{i=1}^{\Omega} e^{-\beta U} = \Omega\, e^{-\beta U}, \\ S &= k_B\left(\beta U + \ln\Omega - \beta U\right) = k_B \ln\Omega. \end{aligned}$$

3.1 First and Second Law of Thermodynamics - Making Sense of Multipliers

The multipliers need thermodynamics for their meaning. The MaxEnt formalism does not by itself provide the physical interpretation or numerical calibration of its multipliers. In

particular, the identification

$$\boxed{\beta = \frac{1}{k_B}\frac{\partial S}{\partial \langle U \rangle} = \frac{1}{k_B T}} \tag{3.11}$$

cannot be derived from MaxEnt alone. It is an empirical correspondence established by classical thermodynamics. We therefore obtain the physical meaning of the multipliers by connecting them to the thermodynamic laws.

From the first and second laws to the fundamental relation. The first law expresses energy conservation:

$$\boxed{dU = \delta Q - \delta W + \mu\, dN,} \tag{3.12}$$

where

- $\delta W = P\, dV$ is the work done *by* the system.
- The symbols δQ and δW emphasize that heat and work are not state functions: their values depend on the path between two states.
- The chemical potential μ is the change in internal energy associated with adding particles at fixed entropy and volume.

The second law, in the form of the Clausius inequality, states

$$\boxed{\delta Q \leq T\, dS,} \tag{3.13}$$

with equality for reversible processes. Combining the two laws for a reversible process gives

$$\boxed{dU = T\, dS - P\, dV + \mu\, dN,} \tag{3.14}$$

and therefore

$$\boxed{\left.\frac{\partial U}{\partial S}\right|_{V,N} = T, \qquad \left.\frac{\partial U}{\partial V}\right|_{S,N} = -P, \qquad \left.\frac{\partial U}{\partial N}\right|_{S,V} = \mu.} \tag{3.15}$$

Rearranging Eq. (3.14) gives the equivalent form

$$\boxed{dS = \frac{1}{T}\, dU + \frac{P}{T}\, dV - \frac{\mu}{T}\, dN,} \tag{3.16}$$

and therefore

$$\boxed{\left.\frac{\partial S}{\partial U}\right|_{V,N} = \frac{1}{T}, \qquad \left.\frac{\partial S}{\partial V}\right|_{U,N} = \frac{P}{T}, \qquad \left.\frac{\partial S}{\partial N}\right|_{U,V} = -\frac{\mu}{T}.} \tag{3.17}$$

3.2 Extensive and Intensive Variables

Extensive and intensive variables and their conjugate pairs.

- A quantity X is extensive if, for two weakly coupled subsystems, $X_{\text{total}} = X_1 + X_2$. Equivalently, for λ identical, non-interacting copies, $X_\lambda = \lambda X_1$.
- A quantity Y is intensive if it does not scale with the number of copies: $Y_\lambda = Y_1$.
- Energy, particle number, volume, and entropy are extensive, while temperature, pressure, and chemical potential are intensive.

This distinction explains why the Lagrange multipliers associated with extensive constraints are intensive. For any observable A with constrained average $\langle A \rangle$, the multiplier is

$$\beta_A \equiv \frac{1}{k_B} \frac{\partial S}{\partial \langle A \rangle}.$$

Because $\langle A \rangle$ and S are both extensive, their scaling factors cancel in the derivative, so β_A is intensive. Thermodynamic variables therefore come in conjugate pairs: an extensive variable and its intensive conjugate.

Additivity of entropy follows from statistical independence. Here, the superscripts (1) and (2) also label the distributions of the two systems, $p_i^{(1)}$ and $p_j^{(2)}$. The partition function factorizes because the joint distribution does: $p_{ij} = p_i^{(1)} p_j^{(2)}$. The entropy is then additive:

$$S_{1+2} = -\sum_{i,j} p_i^{(1)} p_j^{(2)} \ln\!\left(p_i^{(1)} p_j^{(2)}\right) = S_1 + S_2.$$

This requires statistical independence, not equilibrium, so the two systems may have different multipliers β_1 and β_2:

$$\boxed{S_{1+2} = k_B \left(\beta_1 \langle U_1 \rangle + \ln Z_1 + \beta_2 \langle U_2 \rangle + \ln Z_2\right).} \tag{3.18}$$

Only when $\beta_1 = \beta_2 = \beta$, that is, when the two systems are in thermal equilibrium, does the joint system become canonical with a single multiplier:

$$\boxed{S_{1+2} = k_B \left[\beta \langle U_{1+2} \rangle + \ln Z_{1+2}\right], \qquad U_{1+2} = U_1 + U_2.} \tag{3.19}$$

Independence holds when the coupling between the systems is weak. For strong coupling, the joint distribution does not factorize, and $S_{1+2} = S_1 + S_2 - I$ with the mutual information $I \geq 0$.

3.3 Energy Fluctuation, Thermodynamic Limit, and Equivalence of Ensembles

Energy fluctuation and the heat capacity. The variance of the internal energy, the energy fluctuation around the controlled mean $\langle U \rangle = \bar{U}$, can be written in familiar thermodynamic quantities. By the chain rule,

$$\mathrm{Var}(U) = -\frac{\partial \langle U \rangle}{\partial \beta} = -\frac{\partial \langle U \rangle}{\partial T} \frac{\partial T}{\partial \beta}.$$

Since $T = 1/(k_B \beta)$, we have $\partial T / \partial \beta = -k_B T^2$. By definition, $\partial \langle U \rangle / \partial T|_{V,N} \equiv C_V$ is the isochoric heat capacity. It is computed at fixed volume and particle number, which are the natural variables of the canonical ensemble. Substituting,

$$\boxed{\mathrm{Var}(U) = k_B T^2 C_V.} \tag{3.20}$$

Why fluctuations vanish in the thermodynamic limit. One might attribute the smallness of fluctuations to the tiny numerical value of k_B in SI units. This is not the reason: the value of k_B only reflects our choice of units. The genuine reason is that C_V and $\langle U \rangle$ are both extensive and scale linearly with the particle number N. Hence

$$\boxed{\frac{\sqrt{\mathrm{Var}(U)}}{\langle U \rangle} \sim \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}},} \tag{3.21}$$

which vanishes as $N \to \infty$. For a macroscopic system, relative energy fluctuations are negligible because of the sheer size of N. For this reason, we write U in place of $\langle U \rangle$ from here on, and likewise for other extensive variables. At large N, the canonical ensemble therefore behaves as the microcanonical one: the fluctuation is negligible, and the relation between energy and temperature is one-to-one.

Physical reading of the fluctuation formula. At higher temperature, β is smaller, so the entropic cost per unit of exchanged energy is smaller. The system then absorbs or releases energy more freely, consistent with the larger fluctuation $k_B T^2 C_V$.

Concavity of $S(U)$ and thermodynamic stability. Applying the concavity result to $A = U$ and using $\mathrm{Var}(U) = k_B T^2 C_V$,

$$\boxed{\frac{\partial^2 S}{\partial U^2} = -\frac{k_B}{\mathrm{Var}(U)} = -\frac{1}{C_V\, T^2}.} \tag{3.22}$$

Concavity of $S(U)$ is therefore equivalent to $C_V > 0$, the thermodynamic stability condition: adding energy to a system must raise its temperature. This property is needed again when we introduce Legendre transformations.

3.4 Zeroth Law of Thermodynamics

Two bodies exchanging energy. Consider two bodies, (1) and (2), in contact so that they can exchange energy. Two points need to be stated precisely:

- The combined system (1) + (2) is genuinely isolated: nothing is exchanged with the outside. The total energy is a hard constraint, $U_1 + U_2 = U_{\mathrm{tot}}$, so whatever one body releases, the other receives. This is a microcanonical condition on the pair as a whole.
- Each body has an entropy function $S_i(U_i, V_i, N_i)$, which is the canonical entropy derived earlier in this chapter. Each body carries the multiplier $\beta_i \equiv k_B^{-1}\, \partial S_i / \partial U_i$ attached to its own energy. The pair is microcanonical, while each body behaves canonically with the other body as its energy reservoir.

Equilibrium requires equal multipliers. The two bodies are weakly coupled, so the combined entropy is additive, $S = S_1 + S_2$. Equilibrium corresponds to the maximum of S with respect to how U_{tot} is shared. Substituting $U_2 = U_{\text{tot}} - U_1$ and differentiating with respect to U_1,

$$\frac{\partial S}{\partial U_1} = \frac{\partial S_1}{\partial U_1} + \frac{\partial S_2}{\partial U_2}\frac{\partial U_2}{\partial U_1} = \frac{\partial S_1}{\partial U_1} - \frac{\partial S_2}{\partial U_2} = 0.$$

Since $\beta_i = k_B^{-1}\,\partial S_i/\partial U_i$, this gives

$$\boxed{\beta_1 = \beta_2, \qquad \text{i.e.} \qquad T_1 = T_2.} \tag{3.23}$$

Direction of energy flow and the zeroth law. Suppose $\beta_1 \neq \beta_2$ initially. A small energy transfer δU_1 into body (1) changes the total entropy by

$$\delta S = k_B(\beta_1 - \beta_2)\,\delta U_1.$$

The second law requires $\delta S \geq 0$, so energy flows into the body with the larger β, that is, the lower T. Energy therefore flows from high temperature to low temperature, as expected. A third body in contact with both must, by the same argument, reach the same β at equilibrium. This transitivity is the zeroth law of thermodynamics.

Volume and particle exchange cannot be separated from energy exchange. Suppose the two bodies can also exchange volume through a movable partition, with $V_1 + V_2 = V_{\text{tot}}$ fixed. Volume exchange does not by itself imply thermal contact: a heat-insulating piston that is free to slide permits volume exchange and forbids heat exchange, so $T_1 \neq T_2$ in general. The multiplier conjugate to volume is $\beta_V \equiv k_B^{-1}\,\partial S/\partial V = P/(k_B T)$. Maximizing $S = S_1(V_1) + S_2(V_2)$ over the split of V_{tot}, with U_1 and U_2 fixed, gives

$$\beta_{V,1} = \beta_{V,2} \qquad \Longrightarrow \qquad \frac{P_1}{T_1} = \frac{P_2}{T_2}.$$

This permits $P_1 \neq P_2$, yet a partition under a genuine pressure imbalance cannot rest, by Newton's second law alone. The flaw is the assumption that U_1 and U_2 stay fixed while the partition moves. A sliding wall does work $P\,dV$ on the side it pushes into, so even a heat-insulating wall exchanges energy through work, $dU = -P\,dV$ for $\delta Q = 0$. If energy and volume can both be exchanged, as for a movable wall that also conducts heat, maximizing $S_1(U_1, V_1) + S_2(U_2, V_2)$ over both splits gives

$$\boxed{T_1 = T_2 \quad \text{and} \quad P_1 = P_2,} \tag{3.24}$$

the physically sensible free-piston equilibrium.

More generally, exchanging volume or particle number is never independent of exchanging energy. Each carries its own energy contribution in the first law ($-P\,dV$ and $\mu\,dN$), so letting either vary freely lets the energy vary too. This is why the multipliers conjugate to volume and particle number always appear entangled with the temperature: their exchange is, at bottom, still routed through energy.

4 Beyond Canonical Ensemble

4.1 Internal Energy Minimization

Two ways to slice the entropy landscape. Consider the entropy $S(U, X)$ as a surface over the (X, U) plane, where X is an extensive variable that can be freely partitioned, such as a volume shared between two bodies. Draw the level curves of S, the curves along which S is constant, like contour lines on a topographic map. The map can be sliced in two ways:

- Entropy maximum at fixed U: walk along a horizontal line, $U = \text{const}$, and find the highest contour it reaches. At that point the line touches the contour without crossing it, so the contour is tangent to the line.
- Energy minimum at fixed S: walk along one contour, $S = \text{const}$, and find its lowest point. There the contour has a horizontal tangent.

Both slices therefore select a point where the level curve of S has a horizontal tangent in the (X, U) plane.

One geometric condition, two descriptions. The gradient

$$\nabla S = \left(\frac{\partial S}{\partial X}, \frac{\partial S}{\partial U} \right)$$

is perpendicular to the level curve. If $\partial S/\partial X = 0$ at a point, the gradient points purely along the U-direction, so the level curve points purely along the X-direction: it is horizontal. A horizontal tangent means $dU/dX|_S = 0$, a stationary point of U along the contour. The implicit function theorem makes this explicit:

$$\boxed{\left.\frac{\partial U}{\partial X}\right|_S = -\frac{\partial S/\partial X\,|_U}{\partial S/\partial U\,|_X}.} \tag{4.1}$$

Since $\partial S/\partial U = 1/T$ is finite and nonzero, the left side vanishes exactly when the numerator vanishes:

$$\boxed{\left.\frac{\partial S}{\partial X}\right|_U = 0 \quad \Longleftrightarrow \quad \left.\frac{\partial U}{\partial X}\right|_S = 0.} \tag{4.2}$$

The two conditions are not coincidentally equal. They describe the same point on the same contour line.

Generality of the argument and the role of stability. This argument does not depend on entropy or energy. For any smooth function $f(x, y)$ that is monotonic in y at fixed x, the points where f is extremal in x at fixed y coincide with the points where y is extremal in x along a level set of f, because both say that the contour is flat. Thermodynamics is the case $f = S$ and $y = U$, with x any extensive variable; monotonicity follows from $\partial S/\partial U = 1/T > 0$. What the thermodynamic stability condition, $\partial^2 S/\partial X^2 < 0$, adds is the type of flat point: a maximum of S at fixed U corresponds to a minimum of U at fixed S.

Which flat point: maximum of S, minimum of U. We differentiate the implicit-function relation once more along a contour of constant S. Using $\partial U/\partial X|_S = -T\,\partial S/\partial X|_U$,

$$\left.\frac{\partial^2 U}{\partial X^2}\right|_S = -T\left.\frac{d}{dX}\right|_S\left(\left.\frac{\partial S}{\partial X}\right|_U\right) - \left.\frac{\partial T}{\partial X}\right|_S\left.\frac{\partial S}{\partial X}\right|_U.$$

At the stationary point, $\partial S/\partial X|_U = 0$ and $\partial U/\partial X|_S = 0$. The second term then vanishes, and in the first term the chain rule reduces to $\partial^2 S/\partial X^2|_U$:

$$\boxed{\left.\frac{\partial^2 U}{\partial X^2}\right|_S = -T\left.\frac{\partial^2 S}{\partial X^2}\right|_U.} \tag{4.3}$$

Since $T > 0$ and $\partial^2 S/\partial X^2 < 0$ by stability, the right side is positive. A maximum of S at fixed U is therefore a minimum of U at fixed S.

4.2 Thermodynamic Potential

Entropy as the microcanonical thermodynamic potential. In the microcanonical ensemble, the entropy $S(U, V, N)$ is a function of state: the values of U, V, and N determine S completely. Differentiating S yields every conjugate intensive variable, and S is the extremized quantity of the ensemble. For these reasons, S is called the thermodynamic potential of the ensemble, and (U, V, N) are its natural variables. A potential $\Psi(x_1, \ldots, x_n)$ has natural variables $(x_1, \ldots, x_n)$ when:

(i) differentiating Ψ with respect to each x_i recovers every conjugate variable: $d\Psi$ is an exact differential in exactly these variables, with nothing missing and nothing required from outside;

(ii) the x_i are precisely the quantities externally fixed or controlled under the physical condition that Ψ describes, that is, the hard constraints of the ensemble, not an arbitrary choice of variables.

The need for a potential with natural variables (N, V, T). Because S is strictly increasing in U, it can be inverted to give $U(S, V, N)$, which is also a function of state. Its derivatives likewise yield T, P, and μ, and U is minimized at fixed (S, V, N). In the canonical ensemble, however, U is no longer a controlled parameter; the temperature is. The entropy is then known only as a consequence of T, not as a coordinate we can vary at will. We therefore need a potential that preserves the full information of the system and uses the natural variables of the ensemble, (N, V, T).

4.3 The Legendre Transformation

Constructing the transform from tangent lines. In this section we denote the slope by ξ instead of p, to avoid confusion with the probability p_i and the pressure P. Let $f(x)$ be a function. At each point x, the tangent line has slope $\xi(x) = f'(x)$ and intercept $f(x) - \xi(x)\,x$. Expressing the intercept as a function of the slope defines the Legendre transform

$$\boxed{g(\xi) = f(x) - \xi\,x, \qquad \xi = f'(x).} \tag{4.4}$$

We say that $g(\xi)$ conserves the information of $f(x)$ if f can be reconstructed from g alone.

Reconstructing f from its transform. Each tangent line, labeled by its slope ξ, is $y = \xi x + g(\xi)$, so the tangent lines form a family with one line per ξ. Fix a point x and consider the vertical gap between this family and the curve,

$$\Delta(\xi, x) = \xi x + g(\xi) - f(x).$$

A function f is convex if $-f$ is concave, that is, if the inequality in the definition of concavity is reversed. If f is convex, every tangent line lies on or below the curve, so $\Delta \leq 0$ for all ξ, with equality exactly for the line tangent at x. If f is concave, every tangent line lies on or above the curve, so $\Delta \geq 0$ with the same equality condition. In either case, the tangent line is a genuine extremum of Δ over ξ, not merely a stationary point:

$$\frac{\partial \Delta}{\partial \xi} = x + g'(\xi) = 0 \quad \Longleftrightarrow \quad x = -g'(\xi).$$

Calling the solution $\xi^*(x)$, we reconstruct f as

$$\boxed{x = -g'(\xi), \qquad f(x) = \xi^*(x)\, x + g\big[\xi^*(x)\big].} \tag{4.5}$$

Hence f and g determine each other, and no information is lost. The transformation trades the variable x for its conjugate $\xi = f'(x)$.

4.4 Multivariable Legendre Transformation

Extending the transform to several variables. Let $f(x_1, \ldots, x_n)$ be a function of $\mathbf{x} = (x_1, \ldots, x_n)$. At each point, the tangent hyperplane has gradient $\boldsymbol{\xi}(\mathbf{x}) = \nabla f(\mathbf{x})$, with components $\xi_i = \partial f / \partial x_i$, and intercept $f - \boldsymbol{\xi} \cdot \mathbf{x}$. The Legendre transform is

$$\boxed{g(\boldsymbol{\xi}) = f(\mathbf{x}) - \boldsymbol{\xi} \cdot \mathbf{x} = f(\mathbf{x}) - \sum_{i=1}^{n} \xi_i x_i.} \tag{4.6}$$

Reconstructing f from its transform. Each tangent hyperplane, labeled by its gradient $\boldsymbol{\xi}$, is $y = \boldsymbol{\xi} \cdot \mathbf{x} + g(\boldsymbol{\xi})$, so the hyperplanes form a family with one member for each $\boldsymbol{\xi} \in \mathbb{R}^n$. Fix a point $\mathbf{x}$ and consider the vertical gap between this family and the surface,

$$\Delta(\boldsymbol{\xi}, \mathbf{x}) = \boldsymbol{\xi} \cdot \mathbf{x} + g(\boldsymbol{\xi}) - f(\mathbf{x}).$$

If f is convex, every tangent hyperplane lies on or below the surface, so $\Delta \leq 0$ for all $\boldsymbol{\xi}$, with equality exactly for the hyperplane tangent at $\mathbf{x}$. If f is concave, $\Delta \geq 0$ with the same equality condition. In either case, that is, when the Hessian of f is semidefinite everywhere, the tangent hyperplane is a genuine extremum of Δ over $\boldsymbol{\xi}$:

$$\frac{\partial \Delta}{\partial \xi_i} = x_i + \frac{\partial g}{\partial \xi_i} = 0 \quad \Longleftrightarrow \quad x_i = -\frac{\partial g}{\partial \xi_i} \qquad (i = 1, \ldots, n).$$

Calling the solution $\boldsymbol{\xi}^*(\mathbf{x})$, we reconstruct f as

$$\boxed{x_i = -\frac{\partial g}{\partial \xi_i}, \qquad f(\mathbf{x}) = \boldsymbol{\xi}^*(\mathbf{x}) \cdot \mathbf{x} + g\big[\boldsymbol{\xi}^*(\mathbf{x})\big].} \tag{4.7}$$

No information is lost: the transformation trades the variables $x_1, \ldots, x_n$ for their conjugates $\xi_i = \partial f / \partial x_i$.

Partial Legendre transforms. In thermodynamics, one often transforms only a subset $I \subseteq \{1, \ldots, n\}$ of the variables and leaves the rest untouched:

$$\boxed{g\big(\{\xi_i\}_{i\in I}, \{x_j\}_{j\notin I}\big) = f(\mathbf{x}) - \sum_{i\in I} \xi_i x_i, \qquad \xi_i = \left.\frac{\partial f}{\partial x_i}\right|_{\mathbf{x}}.} \tag{4.8}$$

The construction above goes through one variable at a time. Each x_i with $i \in I$ is swapped for its conjugate ξ_i, and each x_j with $j \notin I$ is held fixed as a spectator.

Convexity of partial transforms. If f is convex, a partial Legendre transform is concave in the transformed variables ξ_i and convex in the untransformed variables x_j. Since $U(S, V, N)$ is convex by thermodynamic stability, $F(T, V, N)$ is concave in T and convex in V, and $H(S, P, N)$ is concave in P and convex in S. This is why F is minimized over V at fixed T, but not over T.

4.5 NVT Ensemble - Helmholtz Free Energy

Helmholtz free energy as the Legendre transform of U. The thermodynamic potential of the NVT ensemble, or canonical ensemble, is the Helmholtz free energy. It is the partial Legendre transform of $U(S, V, N)$ that trades S for its conjugate $T = \partial U / \partial S|_{V,N}$:

$$F(N, V, T) = U - \left.\frac{\partial U}{\partial S}\right|_{V,N} S,$$

that is,

$$\boxed{F(N, V, T) = U - TS.} \tag{4.9}$$

Conjugate variables of F. We check that F generates every conjugate variable. Since U depends on S, and S in turn depends on T, V, and N, each derivative of F must include the contribution of S:

$$\begin{aligned}
\left.\frac{\partial F}{\partial T}\right|_{V,N} &= \frac{\partial U}{\partial S}\frac{\partial S}{\partial T} - S - T\frac{\partial S}{\partial T} = -S, \\
\left.\frac{\partial F}{\partial V}\right|_{T,N} &= \frac{\partial U}{\partial V} + \frac{\partial U}{\partial S}\frac{\partial S}{\partial V} - T\frac{\partial S}{\partial V} = -P + T\frac{\partial S}{\partial V} - T\frac{\partial S}{\partial V} = -P, \\
\left.\frac{\partial F}{\partial N}\right|_{T,V} &= \frac{\partial U}{\partial N} + \frac{\partial U}{\partial S}\frac{\partial S}{\partial N} - T\frac{\partial S}{\partial N} = \mu + T\frac{\partial S}{\partial N} - T\frac{\partial S}{\partial N} = \mu.
\end{aligned}$$

Hence

$$\boxed{\left.\frac{\partial F}{\partial T}\right|_{V,N} = -S, \qquad \left.\frac{\partial F}{\partial V}\right|_{T,N} = -P, \qquad \left.\frac{\partial F}{\partial N}\right|_{T,V} = \mu.} \tag{4.10}$$

F behaves much like U: it has the same dimension and yields the same P and μ, while T has become an argument and $-S$ its conjugate.

Minimization of F from entropy maximization. The same result follows from MaxEnt applied to the combined system. Let the system, at fixed V and N, exchange energy with a reservoir at temperature T, and treat the pair as isolated, as in the zeroth-law setup. The reservoir gains the energy that the system loses, so $dS_{\text{res}} = -dU/T$, and the total entropy changes by

$$\boxed{dS_{\text{tot}} = dS - \frac{dU}{T} = -\frac{dF}{T} \geq 0.} \tag{4.11}$$

Maximizing the total entropy is therefore equivalent to minimizing F of the system alone. The factor $-TS$ in F accounts for the reservoir.

Free energy and the partition function. Substituting $S = k_B(\beta U + \ln Z)$ and $\beta = 1/(k_B T)$ into $F = U - TS$ gives

$$F = U - \frac{1}{k_B \beta}\, k_B\, (\beta U + \ln Z) = U - U - \frac{\ln Z}{\beta},$$

so

$$\boxed{F = -k_B T \ln Z.} \tag{4.12}$$

4.6 NPS Ensemble - Enthalpy

Enthalpy as the Legendre transform of U in the volume. For some systems, it is more natural to control the pressure than the volume, because pressure is intensive and does not depend on system size. We label this the NPS ensemble, after the natural variables of H. Its thermodynamic potential is the enthalpy, the partial Legendre transform of $U(S, V, N)$ that trades V for its conjugate $-P = \partial U/\partial V|_{S,N}$:

$$H(N, P, S) = U - \left.\frac{\partial U}{\partial V}\right|_{S,N} V,$$

that is,

$$\boxed{H(N, P, S) = U + PV.} \tag{4.13}$$

It generates every conjugate variable:

$$\boxed{\left.\frac{\partial H}{\partial S}\right|_{P,N} = T, \qquad \left.\frac{\partial H}{\partial P}\right|_{S,N} = V, \qquad \left.\frac{\partial H}{\partial N}\right|_{S,P} = \mu.} \tag{4.14}$$

Enthalpy as exchanged heat, and its minimization. By the first law, at constant (N, P),

$$\boxed{\delta Q = dU + P\,dV = dH.} \tag{4.15}$$

The Clausius inequality, $\delta Q \leq T\,dS$, then gives $dH \leq T\,dS$, and at constant entropy

$$\boxed{dH \leq 0.} \tag{4.16}$$

Any spontaneous process at fixed (N, P, S) therefore proceeds toward lower H and halts when H reaches its minimum, which recovers the equilibrium condition established earlier. Most processes are not isentropic, however, so enthalpy is mainly used as the heat exchanged at fixed pressure. For example, it measures the heat of a phase transition in chemistry.

Labels and conventional names of ensembles. From here on, we label each ensemble by the natural variables of its thermodynamic potential, such as NVT for F and NPS for H. The conventional name is given alongside. The conventional label of an ensemble lists the quantities held fixed as constraints. For NPS, the conventional name is the isoenthalpic-isobaric ensemble, NPH, which fixes N, P, and the enthalpy H. It relates to $H(N, P, S)$ as the microcanonical ensemble NVU relates to $U(S, V, N)$: the same state, described either by the fixed energy-like quantity or by the entropy. The ensembles in this chapter and the next are:

Natural variables	Potential	Conventional name
(S, V, N)	U	microcanonical (NVU)
(T, V, N)	F	canonical (NVT)
(S, P, N)	H	isoenthalpic-isobaric (NPH)
(T, V, μ)	grand potential	grand canonical (μVT)
(T, P, N)	G	isothermal-isobaric (NPT)

5 The Grand Canonical Ensemble

Adding a second constraint: particle exchange. Likewise, the grand canonical ensemble adds a second constrained average, the particle number, so the system exchanges both energy and particles with its surroundings. Here, β and γ denote the multipliers of the constraints $\langle U\rangle = \bar{U}$ and $\langle N\rangle = \bar{N}$, and U_i and N_i denote the energy and particle number of microstate i. The stationarity conditions of the Lagrangian are

$$\frac{\partial \mathcal{L}}{\partial p_i} = -\ln p_i - 1 - \alpha - \beta U_i - \gamma N_i = 0, \qquad \frac{\partial \mathcal{L}}{\partial \alpha} = -\Big(\sum_i p_i - 1\Big) = 0,$$
$$\frac{\partial \mathcal{L}}{\partial \beta} = -\Big(\sum_i U_i p_i - \bar{U}\Big) = 0, \qquad \frac{\partial \mathcal{L}}{\partial \gamma} = -\Big(\sum_i N_i p_i - \bar{N}\Big) = 0.$$

The grand canonical distribution. The first condition gives $p_i = e^{-1-\alpha-\beta U_i-\gamma N_i}$. Normalization fixes α, and the remaining conditions fix the averages:

$$e^{1+\alpha} = \sum_i e^{-\beta U_i-\gamma N_i}, \qquad \bar{U} = \frac{\sum_i U_i\, e^{-\beta U_i-\gamma N_i}}{\sum_i e^{-\beta U_i-\gamma N_i}}, \qquad \bar{N} = \frac{\sum_i N_i\, e^{-\beta U_i-\gamma N_i}}{\sum_i e^{-\beta U_i-\gamma N_i}}.$$

As β is the control parameter of the canonical ensemble, β and γ are the free parameters here, and α is a normalizing factor that depends on both. Defining the grand canonical partition function

$$\boxed{\Xi(\beta,\gamma) \equiv \sum_i e^{-\beta U_i-\gamma N_i},} \tag{5.1}$$

the probability of a microstate with energy U_i and N_i particles is

$$\boxed{p_i = \frac{e^{-\beta U_i-\gamma N_i}}{\Xi(\beta,\gamma)}.} \tag{5.2}$$

Identifying γ with the chemical potential. The entropy of this distribution is

$$\frac{S}{k_B} = -\sum_i p_i \ln p_i = \beta\langle U\rangle + \gamma\langle N\rangle + \ln \Xi.$$

Differentiating and using $\partial \ln \Xi/\partial\beta = -\langle U\rangle$ and $\partial \ln \Xi/\partial\gamma = -\langle N\rangle$, the terms from the implicit dependence cancel, as in the canonical case:

$$dS = k_B\left(\beta\, d\langle U\rangle + \gamma\, d\langle N\rangle\right), \qquad \text{so} \qquad \gamma = \frac{1}{k_B}\frac{\partial S}{\partial \langle N\rangle}\bigg|_{\langle U\rangle, V}.$$

The first law gives $\partial S/\partial N|_{U,V} = -\mu/T$, hence $\gamma = -\beta\mu$. Like $\beta = 1/(k_B T)$, this is the correspondence with classical thermodynamics. From here on, we use (β,μ) in place of (β,γ). The results become

$$\boxed{\Xi(\beta,\mu) = \sum_i e^{-\beta(U_i-\mu N_i)}, \qquad p_i = \frac{e^{-\beta(U_i-\mu N_i)}}{\Xi(\beta,\mu)},} \tag{5.3}$$

and the entropy is

$$\boxed{S = k_B\left[\beta\left(\langle U\rangle - \mu\langle N\rangle\right) + \ln \Xi\right].} \tag{5.4}$$

Energy and particle number are correlated. Exchanging particles also exchanges energy, as the zeroth-law discussion showed. We therefore expect $\langle U\rangle$ alone to be a poor function of β at fixed μ. To see this, we define the covariance of two observables X and Y,

$$\boxed{\mathrm{Cov}(X,Y) \equiv \langle XY\rangle - \langle X\rangle\langle Y\rangle, \qquad \mathrm{Var}(X) = \mathrm{Cov}(X,X).} \tag{5.5}$$

Let $w_i = U_i - \mu N_i$. For any observable X, at fixed V and μ,

$$\begin{aligned}
\left.\frac{\partial\langle X\rangle}{\partial\beta}\right|_{V,\mu} &= \frac{\partial}{\partial\beta}\left(\frac{\sum_i X_i e^{-\beta w_i}}{\Xi}\right) \\
&= -\frac{\sum_i X_i w_i e^{-\beta w_i}}{\Xi} + \frac{\left(\sum_i X_i e^{-\beta w_i}\right)\left(\sum_j w_j e^{-\beta w_j}\right)}{\Xi^2} \\
&= -\langle Xw\rangle + \langle X\rangle\langle w\rangle = -\mathrm{Cov}(X,w).
\end{aligned}$$

For $X = U$ and $X = w$, this gives

$$\boxed{\left.\frac{\partial\langle U\rangle}{\partial\beta}\right|_{V,\mu} = -\mathrm{Var}(U) + \mu\,\mathrm{Cov}(U,N).} \tag{5.6}$$

$$\boxed{\left.\frac{\partial}{\partial\beta}\left(\langle U\rangle - \mu\langle N\rangle\right)\right|_{V,\mu} = -\mathrm{Var}(U-\mu N).} \tag{5.7}$$

The monotonicity of $\langle U\rangle(\beta)$ depends on the sign of $\mu\,\mathrm{Cov}(U,N)$. The combination $\langle U\rangle - \mu\langle N\rangle$ always decreases with β, like $\langle U\rangle$ in the canonical ensemble. It follows from Ξ in the same way:

$$\boxed{\langle U\rangle - \mu\langle N\rangle = -\left.\frac{\partial \ln\Xi}{\partial\beta}\right|_{V,\mu}.} \tag{5.8}$$

Uniqueness of the multipliers. The multipliers (β,γ) are determined uniquely by the constrained means $(\langle U\rangle,\langle N\rangle)$ whenever the map between them is injective. Since $\langle U\rangle = -\partial\ln\Xi/\partial\beta$ and $\langle N\rangle = -\partial\ln\Xi/\partial\gamma$, the Jacobian of this map is minus the Hessian of $\ln\Xi$. By the covariance formula above, that Hessian is the covariance matrix of (U,N):

$$\boxed{\nabla^2_{(\beta,\gamma)}\ln\Xi = \begin{pmatrix}\mathrm{Var}(U) & \mathrm{Cov}(U,N)\\ \mathrm{Cov}(U,N) & \mathrm{Var}(N)\end{pmatrix} \equiv C.} \tag{5.9}$$

A covariance matrix is positive semidefinite. It is positive definite unless some combination $aU_i + bN_i$ takes the same value in every accessible microstate. In the generic case, $\ln\Xi$ is strictly convex, so its gradient map is injective. Each interior pair $(\bar U,\bar N)$ therefore corresponds to exactly one pair (β,γ). This extends the uniqueness argument of the canonical ensemble to two constraints.

5.1 Particle Number Fluctuation, Thermodynamic Limit, and Equivalence of Ensembles (again)

Particle number fluctuation. The particle number fluctuation is the variance of N. At fixed β and V, the weights depend on μ through $e^{\beta\mu N_i}$, so differentiating the mean gives

$$\left.\frac{\partial\langle N\rangle}{\partial\mu}\right|_{T,V} = \beta\left(\langle N^2\rangle - \langle N\rangle^2\right) = \beta\,\mathrm{Var}(N).$$

Hence

$$\boxed{\mathrm{Var}(N) = k_B T \left.\frac{\partial\langle N\rangle}{\partial\mu}\right|_{T,V}.} \tag{5.10}$$

The thermodynamic limit. The derivative of an extensive variable with respect to an intensive one is extensive, so $\mathrm{Var}(N) \sim N$ and

$$\boxed{\frac{\sqrt{\mathrm{Var}(N)}}{\langle N\rangle} \sim \frac{1}{\sqrt{N}}.} \tag{5.11}$$

As for the energy in the canonical ensemble, particle fluctuations are negligible because of the sheer size of N. We therefore write N in place of $\langle N\rangle$, and U in place of $\langle U\rangle$, from here on. For a very large number of particles, the grand canonical ensemble converges to the microcanonical one. We can then use the microcanonical ensemble to solve idealized versions of realistic thermodynamic systems analytically.

5.2 μVT Ensemble - Grand Potential

The grand potential. The natural variables of the grand canonical ensemble are (μ, V, T), hence the label μVT ensemble. As in the canonical ensemble, T controls the internal energy, and the chemical potential μ likewise controls the particle number. The thermodynamic potential is the grand potential, the partial Legendre transform of $U(S, V, N)$ in S and N:

$$\Phi(\mu, V, T) = U - \left.\frac{\partial U}{\partial S}\right|_{V,N} S - \left.\frac{\partial U}{\partial N}\right|_{S,V} N,$$

that is,

$$\boxed{\Phi(\mu, V, T) = U - TS - \mu N = F - \mu N.} \tag{5.12}$$

Conjugate variables of Φ. From $dU = T\,dS - P\,dV + \mu\,dN$,

$$d\Phi = dU - T\,dS - S\,dT - \mu\,dN - N\,d\mu = -S\,dT - P\,dV - N\,d\mu,$$

so

$$\boxed{\left.\frac{\partial\Phi}{\partial T}\right|_{V,\mu} = -S, \qquad \left.\frac{\partial\Phi}{\partial V}\right|_{T,\mu} = -P, \qquad \left.\frac{\partial\Phi}{\partial\mu}\right|_{T,V} = -N.} \tag{5.13}$$

Minimization of Φ at fixed (μ, V, T). At fixed V we have $\delta W = 0$, so the first law gives $\delta Q = dU - \mu\, dN$. The Clausius inequality, $\delta Q \le T\, dS$, then gives $dU - \mu\, dN \le T\, dS$, and at fixed μ and T

$$\boxed{d\Phi = dU - \mu\, dN - T\, dS \le 0.} \tag{5.14}$$

Any spontaneous process at fixed (μ, V, T) therefore proceeds toward lower Φ and halts when Φ reaches its minimum.

Grand potential and the partition function. Using $S = k_B[\beta(U - \mu N) + \ln \Xi]$ and $\beta = 1/(k_B T)$,

$$\Phi = U - \mu N - TS = (U - \mu N) - \frac{1}{k_B \beta}\, k_B\, [\beta(U - \mu N) + \ln \Xi] = -\frac{\ln \Xi}{\beta},$$

so

$$\boxed{\Phi = -k_B T \ln \Xi.} \tag{5.15}$$

5.3 NPT Ensemble - Gibbs Free Energy

The Gibbs free energy. The NPT ensemble, conventionally the isothermal-isobaric ensemble, is the one most used in experiments: the particle number N, the pressure P, and the temperature T are controlled, and the system exchanges both energy and volume with its surroundings. Like the μVT ensemble, it has two intensive natural variables, here P and T. Its thermodynamic potential is the Gibbs free energy, the partial Legendre transform of $U(S, V, N)$ in S and V:

$$G(T, P, N) = U - \left.\frac{\partial U}{\partial S}\right|_{V,N} S - \left.\frac{\partial U}{\partial V}\right|_{S,N} V,$$

that is,

$$\boxed{G(T, P, N) = U - TS + PV = F + PV = H - TS.} \tag{5.16}$$

Conjugate variables of G. From $dG = -S\, dT + V\, dP + \mu\, dN$,

$$\boxed{\left.\frac{\partial G}{\partial T}\right|_{P,N} = -S, \qquad \left.\frac{\partial G}{\partial P}\right|_{T,N} = V, \qquad \left.\frac{\partial G}{\partial N}\right|_{T,P} = \mu.} \tag{5.17}$$

Minimization of G at fixed (N, P, T). At fixed N, the first law gives $\delta Q = dU + P\, dV$, and the Clausius inequality gives $dU + P\, dV \le T\, dS$. At fixed P and T,

$$\boxed{dG = dU - T\, dS + P\, dV \le 0.} \tag{5.18}$$

Any spontaneous process at fixed (N, P, T) therefore proceeds toward lower G and halts when G reaches its minimum.

The NPT ensemble from MaxEnt. The NPT ensemble arises from the same principle as the others. Constrain the average volume, $\langle V\rangle = \bar{V}$, with a multiplier β_V, and let the microstates include the volume, with a sum over V replaced by an integral if V is continuous. The general solution gives $p_i \propto e^{-\beta U_i - \beta_V V}$. As in the earlier cases, $\beta_V = \frac{1}{k_B}\,\partial S/\partial\langle V\rangle = \beta P$, because $\partial S/\partial V|_{U,N} = P/T$. With the partition function

$$\Gamma(\beta, P) = \sum_i e^{-\beta(U_i + PV_i)},$$

the entropy is $S = k_B[\beta(\langle U\rangle + P\langle V\rangle) + \ln\Gamma]$. Solving for the potential gives

$$\boxed{G = U + PV - TS = -k_B T \ln\Gamma.} \tag{5.19}$$

All five ensembles of this chapter therefore follow from entropy maximization under different sets of constraints.

5.4 Euler's Homogeneous Function Theorem

Extensive variables as functions of other variables. We have classified variables as extensive or intensive, but not yet studied how one extensive variable depends on the others. Euler's (homogeneous function) theorem is a general calculus statement for functions with a scaling property. A function $f(x_1, \ldots, x_n)$ is homogeneous of degree k if

$$\boxed{f(\lambda x_1, \ldots, \lambda x_n) = \lambda^k f(x_1, \ldots, x_n).} \tag{5.20}$$

Euler's theorem. If f is differentiable on $\mathbb{R}^n$ and homogeneous of degree k, then

$$\boxed{\sum_{i=1}^{n} x_i \frac{\partial f}{\partial x_i} = k\, f(x_1, \ldots, x_n).} \tag{5.21}$$

Proof of the theorem. Differentiate the defining relation with respect to λ. With $y_i = \lambda x_i$, the chain rule gives

$$\sum_{i=1}^{n} \left.\frac{\partial f}{\partial y_i}\right|_{y=\lambda x} x_i = k\lambda^{k-1} f(x_1, \ldots, x_n).$$

Setting $\lambda = 1$ yields the theorem.

Application to the internal energy. The internal energy is homogeneous of degree one:

$$U(\lambda S, \lambda V, \lambda N) = \lambda\, U(S, V, N).$$

Euler's theorem then gives

$$\boxed{U = \left.\frac{\partial U}{\partial S}\right|_{V,N} S + \left.\frac{\partial U}{\partial V}\right|_{S,N} V + \left.\frac{\partial U}{\partial N}\right|_{S,V} N = TS - PV + \mu N.} \tag{5.22}$$

This is a relation between values, not a statement about the variables of U. The natural variables of U remain (S, V, N), and (T, P, μ) are functions of them, computed from the derivatives of U. They cannot be treated as variables of U: scaling the system changes U, but leaves the intensive variables (T, P, μ) unchanged.

Potentials with fewer extensive variables. In the NPT ensemble, N is the only extensive natural variable, so scaling the system scales G with N alone:

$$G(\lambda N, P, T) = \lambda\, G(N, P, T).$$

Euler's theorem then gives $G = \mu N$ for a single-component system, the formula commonly used in chemistry. The same argument applies to every potential, and we obtain each one by subtracting its transformed terms from $U = TS - PV + \mu N$:

$$\begin{aligned} F &= U - TS = -PV + \mu N, & H &= U + PV = TS + \mu N, \\ \Phi &= U - TS - \mu N = -PV, & G &= U - TS + PV = \mu N. \end{aligned}$$

Hence

$$\boxed{\begin{aligned} F(N, V, T) &= -PV + \mu N, & H(N, P, S) &= TS + \mu N, \\ \Phi(\mu, V, T) &= -PV, & G(N, P, T) &= \mu N. \end{aligned}} \tag{5.23}$$

Ensembles with one intensive natural variable, NVT and NPS, have a potential that depends on two extensive variables. Ensembles with two intensive natural variables, μVT and NPT, have a potential that depends on one extensive variable only.

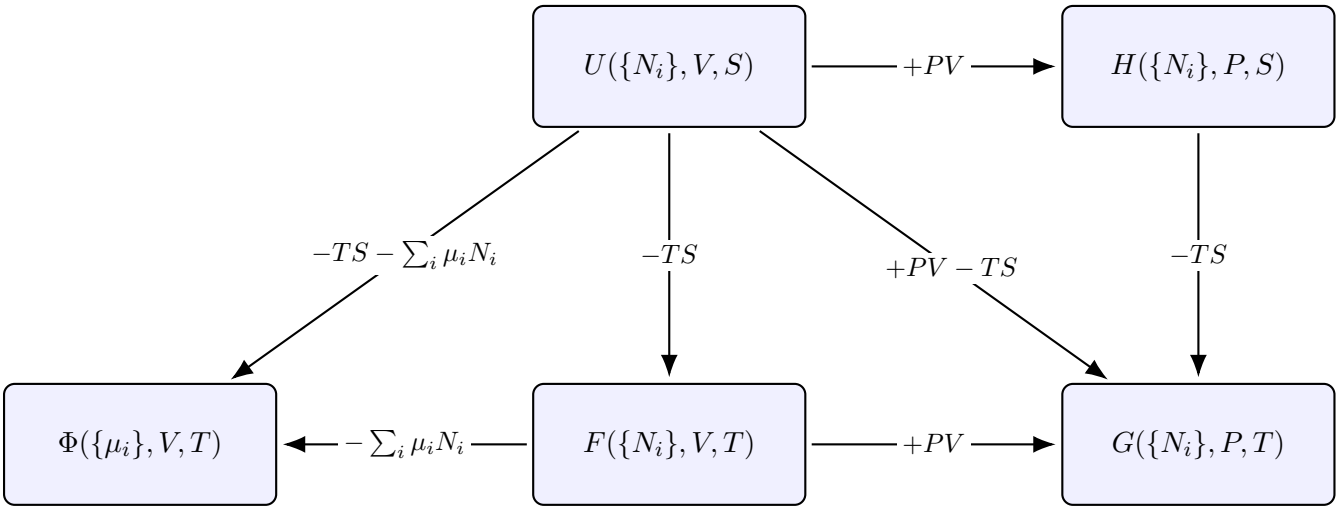
5.5 Multi-Component System

Generalization to several species. So far, we have treated a single species, but realistic systems contain several. For species i, we introduce the particle number N_i and the chemical potential μ_i. The pairs (μ_i, N_i) are conjugate, like the (μ, N) we have used, and each species adds one degree of freedom. From here on, every term μN in a Legendre transformation is replaced by $\sum_i \mu_i N_i$. Euler's theorem becomes

$$\boxed{U = TS - PV + \sum_i \mu_i N_i,} \tag{5.24}$$

and all earlier results carry over.

Relations between the potentials. The potentials are connected by Legendre transformations:



Summary of the ensembles. The table lists, for each ensemble, which variables are fixed and which are obtained by differentiating the potential:

Ensemble	$\{N_i\}VS$	$\{N_i\}VT$	$\{N_i\}PS$	$\{\mu_i\}VT$	$\{N_i\}PT$
Conventional name	micro-canonical	canonical	isoenthalpic-isobaric	grand canonical	isothermal-isobaric
Thermodynamic potential	U	F	H	Φ	G
N_i	fixed	fixed	fixed	$-\frac{\partial \Phi}{\partial \mu_i}$	fixed
V	fixed	fixed	$\frac{\partial H}{\partial P}$	fixed	$\frac{\partial G}{\partial P}$
S	fixed	$-\frac{\partial F}{\partial T}$	fixed	$-\frac{\partial \Phi}{\partial T}$	$-\frac{\partial G}{\partial T}$
μ_i	$\frac{\partial U}{\partial N_i}$	$\frac{\partial F}{\partial N_i}$	$\frac{\partial H}{\partial N_i}$	fixed	$\frac{\partial G}{\partial N_i}$
P	$-\frac{\partial U}{\partial V}$	$-\frac{\partial F}{\partial V}$	fixed	$-\frac{\partial \Phi}{\partial V}$	fixed
T	$\frac{\partial U}{\partial S}$	fixed	$\frac{\partial H}{\partial S}$	fixed	fixed

5.6 Mixing Entropy and the Gibbs Paradox

Setting and the two properties of the entropy. Let $X = (U, V, N_1, N_2)$ denote the extensive variables and $S(X) = k_B \ln \Omega(X)$ the microcanonical entropy. We use two properties, both understood in the thermodynamic limit.

- **(P1) Superadditivity.** $S(X + Y) \geq S(X) + S(Y)$. Place two independent systems with variables X and Y next to each other. Each pair of microstates, one from each system, is a microstate of the combined system with variables $X + Y$, and distinct pairs give distinct microstates. Hence $\Omega(X + Y) \geq \Omega(X)\Omega(Y)$, and taking $k_B \ln$ gives (P1).
- **(P2) Homogeneity.** $S(\lambda X) = \lambda S(X)$ for $\lambda > 0$. This is extensivity. For identical particles it holds only if Ω counts indistinguishable particles, that is, with the factor $1/N!$ included.

Concavity of the entropy along the composition line. For $\lambda \in [0, 1]$, (P1) and (P2) give

$$S\big(\lambda X + (1-\lambda)Y\big) \;\geq\; S(\lambda X) + S\big((1-\lambda)Y\big) = \lambda S(X) + (1-\lambda)S(Y),$$

so S is concave in X:

$$\boxed{S\big(\lambda X + (1-\lambda)Y\big) \;\geq\; \lambda S(X) + (1-\lambda)S(Y).} \tag{5.25}$$

Now fix the total energy $2U$, the volume $2V$, and the total particle number $2N$. The only remaining variable is the composition, which we label by the number t of species-2 particles:

$$g(t) \equiv S(2U,\, 2V,\, 2N - t,\, t), \qquad t \in [0, 2N].$$

The map $t \mapsto (2U, 2V, 2N - t, t)$ is affine, and a concave function composed with an affine map is concave, so g is concave. Three points on this line matter:

$$\begin{aligned} g(0) &= S(2U, 2V, 2N, 0) && \text{(all species 1)},\\ g(2N) &= S(2U, 2V, 0, 2N) && \text{(all species 2)},\\ g(N) &= S(2U, 2V, N, N) && \text{(half and half)}. \end{aligned}$$

Since $(N, N) = \frac{1}{2}(2N, 0) + \frac{1}{2}(0, 2N)$, the half-and-half state is the midpoint of the two pure states.

Entropy of mixing for distinguishable species. Take the two systems before the partition is removed, $X = (U, V, N, 0)$ and $Y = (U, V, 0, N)$, and define the entropy of mixing

$$\boxed{\Delta S_{\text{mix}} \equiv S(2U, 2V, N, N) - S(U, V, N, 0) - S(U, V, 0, N) = g(N) - S(U, V, N, 0) - S(U, V, 0, N).} \tag{5.26}$$

Step 1: concavity at the midpoint. Applying the concavity inequality along the line with $\lambda = \frac{1}{2}$,

$$g(N) \;\geq\; \tfrac{1}{2}\, g(0) + \tfrac{1}{2}\, g(2N).$$

The right-hand side is the height of the chord joining the two end points, at its midpoint, so the graph of g does not lie below its chord.

Step 2: homogeneity at the end points. By (P2) with $\lambda = 2$,

$$g(0) = S(2U, 2V, 2N, 0) = 2\,S(U, V, N, 0), \qquad g(2N) = S(2U, 2V, 0, 2N) = 2\,S(U, V, 0, N).$$

Inserting these into the midpoint inequality gives $g(N) \geq S(U, V, N, 0) + S(U, V, 0, N)$, that is,

$$\boxed{\Delta S_{\text{mix}} \geq 0.} \tag{5.27}$$

The same result follows directly from (P1), since $X + Y = (2U, 2V, N, N)$.

When equality holds. Equality in the midpoint inequality means that $g(N)$ lies on the chord. A concave function that touches its chord at an interior point equals the chord everywhere: if some $g(t_0)$ were strictly above the chord, concavity would put $g(N)$ strictly above it too. Therefore

$$\boxed{\Delta S_{\text{mix}} = 0 \iff g \text{ is affine on } [0, 2N], \qquad \Delta S_{\text{mix}} > 0 \iff g \text{ is not affine.}} \tag{5.28}$$

In particular, $\Delta S_{\text{mix}} > 0$ whenever S is strictly concave along the composition line.

Identical particles and the paradox. For identical particles, take $X = Y = (U, V, N)$. By (P2),

$$S(2U, 2V, 2N) = 2S(U, V, N) = S(X) + S(Y),$$

so $\Delta S_{\text{mix}} = 0$ exactly. In terms of g: if the two "species" are physically identical, S depends only on $N_1 + N_2$, so g is constant, hence affine, and the equality case applies. The difference between the two cases is whether the composition is a genuine variable of S. For distinguishable species, g is not affine, and removing the partition increases the entropy by the strictly positive amount ΔS_{mix}. For identical particles, g is constant and $\Delta S_{\text{mix}} = 0$. No continuity is expected between the two: they differ in what the counting Ω treats as distinct, which is why (P2) holds for identical particles only with the $1/N!$ included.

5.7 Gibbs-Duhem Relation

The intensive variables are not independent. The Gibbs-Duhem relation states that the intensive variables $(T, P, \{\mu_i\})$ are not independent. Write the extensive variables as $x = (S, V, \{N_i\})$ and their conjugates as $\xi = (T, -P, \{\mu_i\})$, using ξ as in the Legendre transformation. Let f be any thermodynamic potential, obtained from U by exchanging the variables in a set A for their conjugates:

$$f = U - \sum_{a \in A} \xi_a x_a,$$

so that $f = U$ for $A = \varnothing$, $f = F$ for $A = \{S\}$, $f = G$ for $A = \{S, V\}$, and so on. Wherever f is differentiable in its natural variables,

$$df = \sum_{a \notin A} \xi_a\, dx_a - \sum_{a \in A} x_a\, d\xi_a.$$

By the Euler relation, $U = \sum_a \xi_a x_a$, so f is also

$$f = \sum_{a \notin A} \xi_a x_a, \qquad \text{hence} \qquad df = \sum_{a \notin A} \xi_a \, dx_a + \sum_{a \notin A} x_a \, d\xi_a.$$

Comparing the two expressions for df gives

$$\boxed{\sum_a x_a \, d\xi_a = S \, dT - V \, dP + \sum_i N_i \, d\mu_i = 0.} \tag{5.29}$$

The result does not depend on the choice of A, so it holds in every ensemble. Of the n intensive variables, only $n - 1$ can be varied independently.

Why no ensemble fixes all the intensive variables. Exchanging all extensive variables for their conjugates, $A = \{S, V, \{N_i\}\}$, gives the potential

$$f = U - TS + PV - \sum_i \mu_i N_i = 0.$$

It vanishes identically by the Euler relation. Such a potential would depend on no extensive variable at all, so nothing would fix the size of the system. A system that exchanges energy, volume, and particles freely, with all of $(T, P, \{\mu_i\})$ held fixed, therefore has no ensemble of its own, and Gibbs-Duhem is the statement of this fact.

Part II

Phase Transtion

6 First-Order Phase Transitions and Phase Diagram

For a given system, a phase is a region of the space of natural variables in which the thermodynamic potential is analytic: infinitely differentiable, with no kinks or divergences. A phase transition is a point where it fails to be analytic. Strictly, such singularities appear only in the thermodynamic limit $N \to \infty$, since the partition function of a finite system is analytic.

We usually study phases at fixed particle numbers, pressure and temperature, for two reasons. Experimentally, these are easy to control. Structurally, the phases inside such a system exchange energy, volume and particles with each other, so at equilibrium $(\{\mu_i\}, P, T)$ must be equal in all phases. The natural choice of potential is then the Gibbs free energy $G(T, P, \{N_i\})$. Because G is extensive,

$$G(T, P, \{N_i\}) = N\, g(T, P, \{x_i\}), \qquad x_i = N_i/N,$$

so the total size N does not affect the analyticity of G; only g does. What matters is the composition $\{x_i\}$, of which $C-1$ are independent for C component. A phase is therefore a region of the intensive variables $(T, P, \{x_i\})$, which span a $(C-1)+2 = C+1$ dimensional space. Collecting every phase in this space gives the phase diagram.

At equilibrium, g is continuous, so a phase transition is never a jump in the potential itself. Adding g as one more axis to the $(C+1)$-dimensional space $(T, P, \{x_i\})$ gives a hypersurface $g(T, P, \{x_i\})$. A first-order transition is a crease (kink) of this hypersurface: g is continuous across it, but its first derivatives jump,

$$s = -\frac{\partial g}{\partial T}, \qquad v = \frac{\partial g}{\partial P},$$

the entropy and volume per particle. Equivalently, $S = -\partial G/\partial T$ and $V = \partial G/\partial P$ are discontinuous, which gives a latent heat $L = T\Delta S$ and a volume change ΔV. Melting and boiling are examples. On the transition locus the coexisting phases share the same $(\{\mu_i\}, P, T)$ but have different s, v and composition, which is why the transition appears as a boundary of a coexistence region in the phase diagram.

Transitions of higher order also exist: the order is the order of the lowest derivative of g that is discontinuous or singular, and for these the hypersurface has no kink and the singularity appears in the curvature. They include the continuous transitions, such as the ferromagnetic Curie point, where response functions like the heat capacity diverge. We defer them to a later section.

6.1 Gibbs Phase Rule

Suppose $\mathcal{P}$ phases of a system with C components coexist. At equilibrium all phases share the same intensive variables $(T, P, \mu_1, \ldots, \mu_C)$, which gives $C+2$ variables. Each phase has its own Gibbs–Duhem relation,

$$S_\alpha\, dT - V_\alpha\, dP + \sum_i N_{i,\alpha}\, d\mu_i = 0, \qquad \alpha = 1, \ldots, \mathcal{P},$$

which gives $\mathcal{P}$ independent constraints on them. The number of degrees of freedom F is the number of variables minus the number of constraints,

$$\boxed{F + \mathcal{P} = C + 2.} \tag{6.1}$$

This is the Gibbs phase rule. It counts only intensive degrees of freedom: the amount of each phase remains free and is not counted. However, this is only applicable if $(T, P, \mu_1, \ldots, \mu_C)$ are the only intensive variables. We will encounter system that depends on the magnetic field $\vec{B}$ of itself, which add another intensive variable to the count.

6.2 Clapeyron Equation and Fixed-Composition (P–T) Phase Diagram

No matter how many components the system has, temperature and pressure always participate. Consider the P–T plane at fixed composition $\{x_i\}$ within the $(C+1)$-dimensional phase diagram. In this slice only (P, T) can vary, so there are 2 degrees of freedom. Two phases 1 and 2 coexist when every chemical potential matches,

$$\mu_i^{(1)}(T, P) = \mu_i^{(2)}(T, P), \qquad i = 1, \ldots, C,$$

which relates P and T and reduces the degrees of freedom from 2 to 1. The coexistence locus is therefore a curve in the plane. Moving along it, the two phases keep equal chemical potentials, so $d\mu_i^{(1)} = d\mu_i^{(2)}$ and, by the Gibbs–Duhem relation at fixed composition,

$$-s_1\, dT + v_1\, dP = -s_2\, dT + v_2\, dP,$$

which gives the Clapeyron equation

$$\boxed{\frac{dP}{dT} = \frac{\Delta s}{\Delta v}}, \qquad \Delta s = s_2 - s_1, \quad \Delta v = v_2 - v_1. \tag{6.2}$$

At coexistence the transition proceeds reversibly at fixed T and P, so the latent heat per particle is $\ell = T\Delta s$, and

$$\boxed{\frac{dP}{dT} = \frac{\ell}{T\,\Delta v}.} \tag{6.3}$$

Now suppose a third phase 3 also coexists. Then

$$\mu_i^{(1)}(T, P) = \mu_i^{(2)}(T, P) = \mu_i^{(3)}(T, P), \qquad i = 1, \ldots, C.$$

Each pair of phases has its own coexistence curve, $\mu^{(1)} = \mu^{(2)}$, $\mu^{(2)} = \mu^{(3)}$ and $\mu^{(1)} = \mu^{(3)}$. A point lying on the first two satisfies $\mu^{(1)} = \mu^{(2)} = \mu^{(3)}$, so it lies on the third as well: only two of the three conditions are independent. Two independent equations in the two unknowns (T, P) leave no degrees of freedom, so three phases can coexist only at an isolated point, the triple point, where the three two-phase curves, the triple curves intersect.

6.3 Unary System

A unary system has only one component, $C = 1$. This is a special case of the fixed-composition slice, since the composition is always $x = 1$. We discuss it because it is the building block of higher-dimensional phase diagrams and introduces some terminology. For $C = 1$ the phase rule reads $F = 3 - \mathcal{P}$, so the number of coexisting phases $\mathcal{P}$ fixes the number of free variables.

Within a single phase ($\mathcal{P} = 1$, $F = 2$), P and T can both be varied freely. The system is said to be in bivariant equilibrium.

When two phases coexist ($\mathcal{P} = 2$, $F = 1$), only one of P and T can be varied freely; the other is then fixed by the Clapeyron relation $P = P(T)$. The system is said to be in univariant equilibrium, and the locus of two-phase coexistence is called a univariant curve.

When three phases coexist ($\mathcal{P} = 3$, $F = 0$), no variable can be changed freely. The system is said to be in invariant equilibrium, and the point where this happens is called an invariant point (the triple point).

7 Binary System

7.1 The X–P–T Diagram

A binary system has two components, $C = 2$, so its composition is described by one independent variable, $X = x_2$ with $x_1 = 1 - X$. The phase diagram is therefore three-dimensional in (P, T, X). One can picture it as stacking the fixed-composition P–T diagrams and varying X bit by bit. The phase rule reads $F = 4 - \mathcal{P}$, and each unary structure is stretched by one dimension.

- Tervariant volume ($\mathcal{P} = 1$, $F = 3$). The bivariant regions of the unary diagram become volumes, where P, T and X can all be varied freely without changing phase. The cross section of such a volume changes with X.
- Bivariant surface ($\mathcal{P} = 2$, $F = 2$). The univariant curves become surfaces. Two coexisting phases generally have different compositions, so each phase has its own bivariant surface, and the two surfaces are joined by tie lines. The region they enclose is the two-phase region, where the overall composition fixes the amount of each phase.
- Univariant curve ($\mathcal{P} = 3$, $F = 1$). The invariant points become curves, along which only one variable can be varied freely. Three bivariant surfaces, one for each pair of phases, meet along each such curve.
- Invariant point ($\mathcal{P} = 4$, $F = 0$). A new kind of invariant point appears, where none of P, T or X can be varied. Every subset of the four phases can coexist near it, so it is where $\binom{4}{3} = 4$ univariant curves, $\binom{4}{2} = 6$ two-phase (bivariant) surfaces and $\binom{4}{1} = 4$ single-phase (tervariant) volumes meet. Therefore, it is also called the quadruple point.

Part III

Classical thermodnynamic

Part IV

Physical Chemistry

Part V

Quantum Statistical Mechanics